Porosity optimization of MoS2-SiO2 TENGs

"What are the possible works that we can do on the porosity structure or other related features to the porosity of the SiO2- MoS2 core shell in order to improve TENG applicabability?" *Generated by LeapSpace*

Best directions to improve TENG performance through porosity in SiO_2–MoS_2 core-shell systems

The most supportable ways to work on porosity are to **(1) introduce controlled porosity or hollow space in the core-shell particle, (2) tune pore size and pore distribution to maximize contact area and charge trapping, and (3) preserve a thin, continuous shell so porosity helps performance without sacrificing humidity shielding and charge retention** [11] [46] [47] [48] [49] [50] [51].

What to do on porosity and related features

Work direction	What to change	Why it should help TENGs
Increase accessible porosity	Create **micro/mesopores**, porous shells, or porous interiors in the core-shell structure	More porosity increases **surface area**, **contact area**, and **charge-trapping sites**, which repeatedly improves TENG output in porous tribolayers and porous composites [46] [47] [48] [49] [50] [52]
Tune pore size and distribution	Move from broad, uncontrolled pores to **uniform, size-controlled pores**	Controlled pore architecture is more likely to give stable charge induction and reproducible output than random porosity; pore size engineering is explicitly highlighted as an interfacial design goal in core-shell mesostructures [53]
Build hierarchical porosity	Combine **macro-/micro-/mesoporous** features rather than one pore scale only	Hierarchical micro/nano structures improve contact area and mechanical resilience in TENGs, and porous systems with multiple size scales have shown strong output gains [46] [54] [55] [56]

Work direction	What to change	Why it should help TENGs
Create hollow or partially voided cores	Introduce **internal void space** or hollow nanoparticles before shell growth	Hollow or porous core-shell designs are valuable because they preserve structural integrity while providing more active interfacial area and diffusion pathways in other energy materials; for MoS_2-based shells, hollow architectures also help accommodate structural constraints [51] [53]
Keep the shell thin and continuous	Avoid overly thick shells; favor a **conformal, uniform coating**	A continuous shell is important for **humidity resistance** and charge stabilization, while excessive thickness can reduce accessible surface area and effective contact unless it is deliberately porous [11] [57] [58]
Add charge-trapping surface chemistry	Pair porosity with **high-dielectric / trap-rich interfaces** or functional groups	Porosity helps only if the surface can store and separate charges; the literature repeatedly ties better TENG output to charge trapping, dielectric tailoring, and interfacial polarization [48] [57] [59] [60] [61]
Maintain hydrophobicity and anti-humidity protection	Use porosity selectively so the surface stays **water-resistant**	For MoS_2-encapsulated SiO_2, the MoS_2 shell is specifically reported to block water penetration and stabilize charge density; pore engineering should not destroy that protection [11] [62] [63] [64]

Most promising research tasks

1. **Porosity optimization study**
   Vary pore volume, pore size, and pore density in a SiO_2–MoS_2 core-shell series, then correlate those variables with output voltage, current, and charge retention. This is directly supported by prior TENG studies showing that **increasing porosity can raise output** through larger contact area and more induced charges [48] [49] [50] [52] [65].

2. **Mesoporous shell engineering**
   Replace a dense shell with a **mesoporous or partially porous shell** so the outer layer still blocks moisture but also provides extra interfacial area. This follows the broader core-shell mesostructure principle that **controlled pore shapes, sizes, and functionalities** are a key design lever [53].

3. **Hollow core-shell construction**
   Make **SiO_2 cores with a void or porous interior** before MoS_2 coating, or create a **porous MoS_2 shell** on SiO_2. Hollow or porous shells are attractive because they can preserve structure while increasing active area and transport pathways [51] [66].

4. **Hierarchical roughness + porosity coupling**
   Combine porosity with **micro/nano surface texturing** so the particle surface has both high area and strong contact electrification. This is consistent with the repeated finding that **surface morphology engineering** is a major route to stronger TENG output [47] [55] [67] [68].

5. **Humidity-resistance trade-off testing**
   Test whether adding porosity hurts the moisture barrier function of the SiO_2/MoS_2 system. This is important because the best-performing MoS_2-encapsulated SiO_2 TENG relies on **a thin hydrophobic MoS_2 shell** to prevent water penetration and reduce charge decay [11].

What not to assume

- **More porosity is not automatically better**: the evidence supports porosity as beneficial in many TENGs, but the optimal level depends on whether the shell remains continuous enough to preserve charge retention and humidity stability [11] [57] [64].
- **Exact pore data for $MoS_2@SiO_2$ TENGs are not directly reported** in the retrieved material, so any specific claim about the best pore size or pore fraction for this exact architecture would be an inference rather than a measured result [11] [46] [50].

Bottom line

For **SiO_2–MoS_2 core-shell TENGs**, the most useful porosity work is to **engineer controlled hierarchical porosity or hollow space while keeping a thin, continuous, hydrophobic shell**. That gives you the main TENG benefits of **larger contact area, more charge-trapping sites, and stronger interfacial polarization** without losing the **humidity shielding and charge-retention** advantages that make this core-shell pair attractive in the first place [11] [48] [49] [52] [57] [59].

Confidence level: Medium — the porosity-to-output link is strongly supported across TENG literature, but the retrieved evidence does not include a direct optimization study for porosity in the exact SiO_2–MoS_2 core-shell TENG architecture.

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